

GridWise

Smart Campus Energy

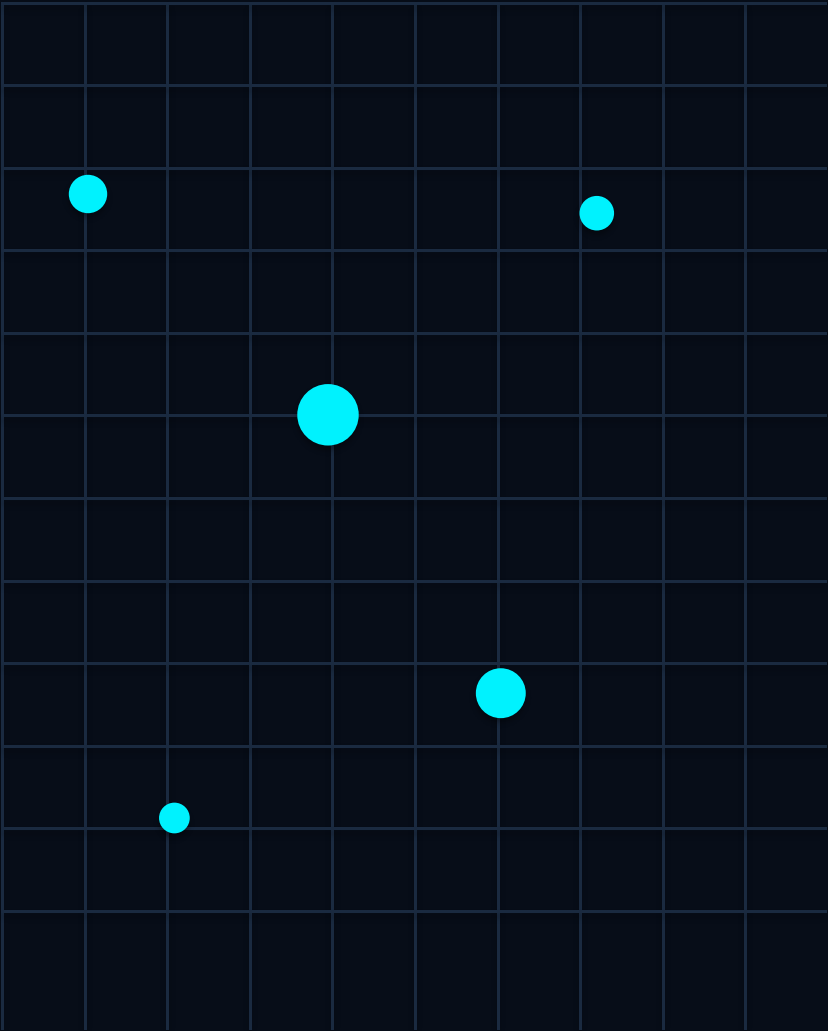
Optimization Service

LLM-Assisted Operator Directive Interpretation
& Mathematically Optimal Linear Programming Dispatch

EWU INNOVATORS

East West University

“Bridging unstructured operator notes with millisecond-level mathematical grid optimization.”



The Real-World Engineering Problem

01 VOLATILITY

5 → 32

BDT / kWh

Time-of-use tariffs can change dramatically across a 24-hour operating horizon.

24h

solar + demand curves

02 HUMAN INPUT

Informal notes

“Solar maintenance at noon”

“Keep 80 kWh reserve”

“Cap feeder imports”

“Cafeteria notice...”

One interface.

Many semantic meanings.

03 ENGINEERING OBJECTIVE

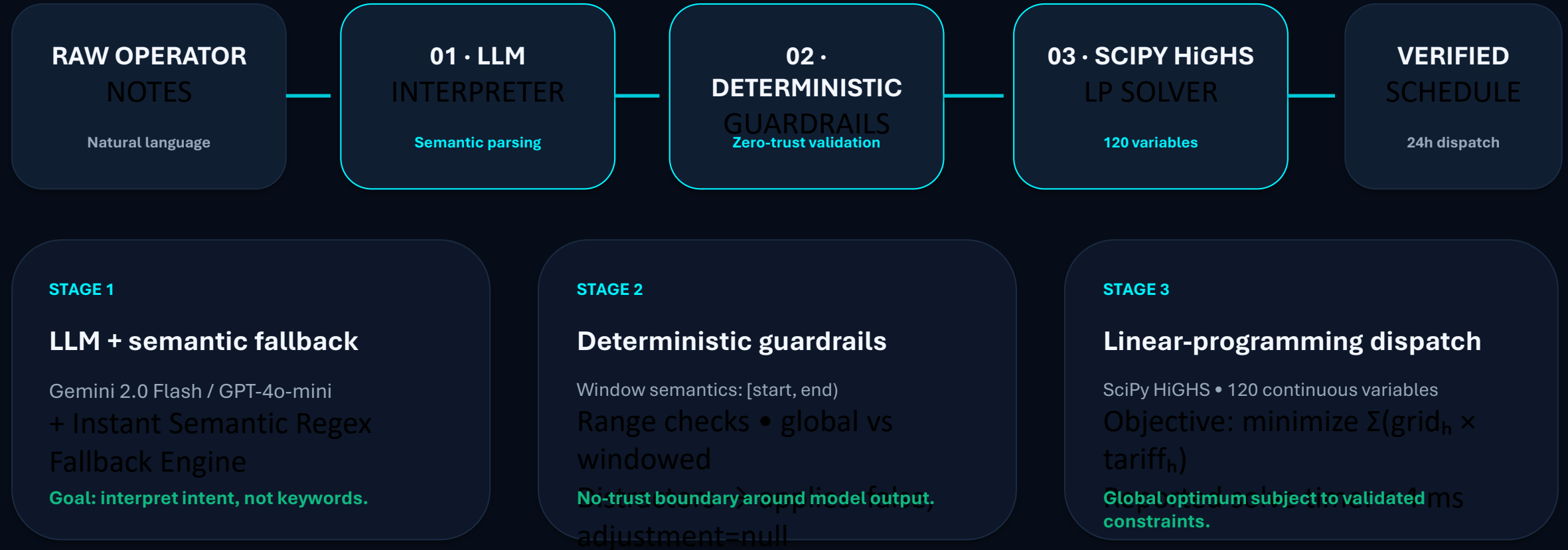
Automate → Validate → Optimize

- ◆ Machine-checkable directives
- ◆ Strict physical constraints
- ◆ Minimum 24h grid electricity cost
- ◆ Headless HTTP API for integration

OUTPUT

Verified hourly dispatch

End-to-End 3-Stage Pipeline Architecture



Robust Directive Coverage & Safety Safeguards

6 SUPPORTED DIRECTIVES

01 solar_reduction

usable factor calculation

02 minimum_battery_reserve

windowed or global reserve

03 no_charge_window

charge rate $\rightarrow 0$

04 no_discharge_window

discharge rate $\rightarrow 0$

05 max_grid_window

transformer intake cap

06 no_op

distractor isolation

P95 LATENCY PROTECTION

3.8 s LLM timeout

Designed to stay inside the ≤ 5.0 s rubric limit.

PHYSICAL INFEASIBILITY

HTTP 400 • structured failure

No crash; mathematically impossible cases are surfaced.

ZERO-LEAK SECURITY

No credentials / secrets

Kept out of repository, logs, and API responses.

Benchmark Validation & Optimization Quality

10 / 10

Public sample cases

100% precision

100.0%

Cost parity

Organizer reference targets

2.9 ms

Avg LP solve

Optimizer runtime

< 4.2 s

Cloud response

≤ 5.0 s rubric limit

HARD PHYSICAL LAWS — CHECKED ON EVERY HOURLY DISPATCH

- | | | |
|----|-----------------------|---|
| 01 | Energy balance | $\text{Grid} + \text{Solar_used} + \text{Discharge} = \text{Demand} + \text{Charge}$ |
| 02 | Solar curtailment | $\text{Solar_used} \leq \text{Effective Solar}$ |
| 03 | Battery bounds | $\text{Active Reserve} \leq \text{Energy} \leq \text{Capacity} + \text{strict rate limits}$ |
| 04 | End-of-day neutrality | $E_{23} = E_{\text{initial}}$ |

VERIFIED DISPATCH

Production Deployment & Immediate Verification

LIVE CLOUD SERVICE

Render deployment

bup-gridwise.onrender.com

READINESS

GET /health

{"status": "ok"} • sub-400 ms

DOCS

GET /docs

OpenAPI / Swagger UI

API

POST /optimize-energy

24h optimization endpoint

CONTAINER FALLBACK

GitHub Container Registry

docker pull

ghcr.io/wahidfarhan/bup_cse_ewu_innovators:latest
multi-arch • CI/CD verified

OPEN-SOURCE REPOSITORY

[github.com/wahidfarhan/
BUP_CSE_EWU_INNOVATORS](https://github.com/wahidfarhan/BUP_CSE_EWU_INNOVATORS)

Production path: API → /health → /docs → /optimize-energy

READY FOR LIVE VERIFICATION